Bootstrapping Vision-Language-Action Control for Cable-Driven Manipulation with Staged Reinforcement Learning

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Bullet points for author approval; results with pending evidence are labeled.

***Abstract*—**

- **Motivation: adapt a pretrained vision-language-action (VLA) system to cable-driven manipulation without collecting new human teleoperation demonstrations for the target robot.**
- **Approach: residual skill acquisition and self-generated initialization, followed by three-stage sparse group-relative reinforcement learning; score unclipped latent actions conditioned on persistent exploration offsets.**
- **Mathematical basis: derive the conditional latent score and binary-group signal probability; prove nondecreasing success in a separable stage model with protected credit; state and test the alignment condition needed by a shared neural controller.**
- **Main result slot: [Author-reported] the latest run exceeds 50% success. Exact score, strict/native definition, checkpoint, and evaluation protocol remain pending; do not substitute the older 41.16% result.**
- **Physical experiment slot: [Pending: hardware results]. State tasks, trial count, and measured success only after completion.**
- **Takeaway to test: staged credit and a consistent action likelihood enable complete manipulation on the target embodiment; bound this claim by matched ablations and observed transfer.**

***Index Terms*—Robot learning, reinforcement learning, manipulation, cable-driven robots.**

I. Introduction

- Problem: pretrained VLA action spaces and training embodiments do not directly match the five control channels of a cable-driven parallel robot (CDPR); complete placement also requires preserving early skills while learning later stages.
- Research question: can a SmolVLA-based residual controller learn empty-gripper pick-and-place through staged reinforcement learning and robot-generated experience, and how much of that behavior survives physical deployment?
- Proposed contribution 1: a documented adaptation pipeline from skill acquisition and consolidation to a single policy executing approach, grasp, lift, carry, and release under one instruction.
- Proposed contribution 2: a mathematical account of conditional latent-action scoring, stage-level signal availability, and achieved-stage credit protection; a sequential-gate improvement proposition and an explicit alignment condition for the shared neural controller. Attribute established ingredients and test the assumptions.
- Proposed contribution 3: matched evaluations, stage/failure analysis, and a physical-robot study **[Pending: completion]**; position the newer $> 50\%$ run using its eventual verified protocol.
- Scope: no newly collected target-robot teleoperation dataset; disclose pretrained data, scripted bridges, oracle-derived historical banks, and human-designed rewards.
- Figure 1 plan: simulation and physical CDPR beside a concise pipeline schematic; hardware images and outcomes remain pending.

II. Related Work

- VLA adaptation: introduce SmolVLA as the backbone [1]; review embodiment/action adaptation and separate frozen-backbone residual training from full VLA fine-tuning.
- Residual learning and autonomous data generation: compare directly with Probe, Learn, Distill (PLD) [2]. Explain differences in learning objective, data lineage, embodiment, and consolidation target; the shared residual-RL/SFT pattern is prior art.
- Group-relative optimization: cite GRPO's origin [3]; add a focused, updated review of robotic VLA RL, sparse sequential credit, and stage-conditioned sampling before final prose.
- Self-imitation and retention: distinguish success-filtered behavior cloning used here from other algorithms called self-imitation learning; retain only the literature needed to explain initialization and interference.
- CDPR learning and sim-to-real: add primary references on cable dynamics, calibration, observation mismatch, and manipulation transfer once the physical setup is fixed.
- Literature comparison plan: supervision source, actor inputs, trainable parameters, full-task resets, and physical evidence. No cross-paper numerical ranking across different task distributions.

III. Problem Formulation

- Define an instruction-conditioned, partially observed manipulation task with ordinary empty-gripper starts and one final placement instruction; specify reset distribution and horizon.

- Actions: normalized $(x, y, z, \text{yaw}, \text{gripper}) \in [-1, 1]^5$, mapped into the CDPR controller. Document chunk length, executed slots, and control frequency from the selected run.
- Observations: overview and wrist images, proprioception, and instruction; enumerate the residual's actual inputs and distinguish them from simulator state used only by rewards/evaluation.
- Primary endpoint: strict full-task success, requiring valid grasp, held lift, intentional release, and native placement without latched carry-slip or wrong-placement events. State temporal ordering and failure termination explicitly.
- Secondary endpoints: native placement, physical grasp, held lift, stage-conditional completion, destination/object breakdowns, and simulator failure rate. These are not interchangeable success definitions.
- Objective: maximize expected strict success on a declared scene distribution. Separate training panels, development evaluation, and a locked final test; no generalization claim follows merely from more repeats of known scenes.

IV. Method

A. VLA-Conditioned Residual Policy

- Specify the mapped prior chunk p, residual input x, and bounded mean $\mu_\theta = \tanh(p + s_r \tanh f_\theta(x, p))$. Preserve both nonlinearities and report the residual scale.
- Identify frozen backbone/action-expert weights, any inherited LoRA, trainable residual weights, and exploration parameters. The latest promoted branch retains the bounded residual; the frozen-reference correction network is a separate experimental arm.
- Explain visual features and language conditioning separately from the action prior. Near-constant prior commands and residual saturation in historical traces motivate prior/feature ablations, not a claim that pretrained action knowledge caused the gain.

B. Bootstrapping and Staged Credit

- Summarize skill acquisition, success-filtered residual SFT, and continuous teacher-chain initialization; provide a lineage diagram identifying policy-generated actions and scripted/oracle assistance.
- Train the complete task from empty starts using ordered approach, pickup, and strict-placement binary milestones; define each predicate and which executed records enter each group.
- Describe group normalization, zero-variance groups, validity masks, and candidate-length/stage weights. Invalid simulator episodes are excluded from learning statistics although counted as evaluation failures; this selection matters mathematically.
- Specify the protected full-task bonus: on achieved approach/pickup milestones, multiply negative terminal advantage by κ; otherwise preserve it. Local settings use bonus weight $\lambda = 1$ and $\kappa = 0$, pending confirmation from the latest run. Placement keeps the signed terminal advantage.
- State the weighted clipped surrogate, frozen-batch advantages, entropy/action penalties, and optimizer. Link each component to the assumptions and limits of Sec. V.
- Explain why prefix competence and complete success differ. Present SFT as an initialization/consolidation component whose benefits require evidence; historical full-task SFT alone did not reliably improve completion.

C. Likelihood Under Clipping and Persistent Exploration

- Record $u_t = \mu_\theta(x_t, p_t) + b_t + \sigma\epsilon_t$ before clipping and execute $a_t = \text{clip}(u_t, -1, 1)$; retain the realized episode-offset contribution b_t and its gate per decision.
- Score $\log \mathcal{N}(u_t; \mu_\theta + b_t, \sigma^2)$ for old/new policy ratios. A parameter-independent offset density cancels when conditioned on its realized value; deterministic clipping belongs to the action-to-environment map.
- Explain the historical mismatch: Gaussian density evaluated at clipped boundaries and independent widened marginals for temporally shared offsets. Include a short derivation or estimator sanity check if space permits.
- Qualification: consistency of the latent score does not prove unbiasedness or convergence of the complete stage-selected, group-normalized, clipped GRPO surrogate.
- Algorithm 1 plan: collect chunks and offsets, observe milestones, form masked stage groups, compute advantages and latent ratios, update trainable parameters, then validate.

V. Mathematical Analysis

A. Correct Likelihood for the Executed Process

- Define $J(\theta) = \mathbb{E}[Y]$ for a fixed stochastic policy, scene distribution and strict outcome. Include controller memory and milestone latches in the augmented history. Distinguish this training objective from deterministic-mean deployment.
- **Proposition A (latent score).** Under fixed dynamics, frozen prior, exogenous offsets and differentiation regularity, factor the augmented trajectory law to derive

$$\nabla J = \mathbb{E}\left[Y \sum_t \nabla \log \pi_\theta(u_t \mid h_t, p_t, b_t) \right]. \qquad (1)$$

- Proof sketch: only policy factors depend on parameters at a fixed history; offset densities cancel and clipping is a deterministic execution map. Include covariance scores if exploration variance is learned. Relate to bounded-action policy-gradient literature [4].
- Scope: per-slot PPO ratios, stage filtering, and invalid-episode exclusion do not inherit unbiased

full-trajectory gradients merely because their log-density is correct.

B. Signal Availability and Protected Stage Credit

- **Lemma B (binary group contrast).** For G independent candidates of one scene with milestone probability p,

$$P_{\text{mixed}} = 1 - p^G - (1-p)^G. \tag{2}$$

- Proof sketch: subtract all-success and all-failure groups. At $G = 8$, probabilities $p = 0.03$ and 0.30 yield approximately 21.6% and 94.2% mixed groups. This predicts usable signal, not guaranteed gradient improvement; saturated milestones also lose contrast.
- Explain the binary-group benchmark $\mathbb{E}[\widehat{g} \mid c] = c_G(p_c)\nabla p_c$ with $c_G > 0$ for full on-policy trajectory scores. Attribute related normalization analysis [5], [6]; stage selection and shared parameters restrict its use here.
- **Proposition C (separable stage model).** In a three-gate model with separate logits $q_k = \text{sigmoid}(\theta_k)$, $J = q_1 q_2 q_3$. With achieved-stage negative-bonus scale $\kappa = 0$, each reached gate's credit has the same sign as its score $X_k - q_k$.
- Proof sketch: achieved gates receive nonnegative credit, failed gates receive nonpositive credit. Nonnegative stage/candidate weights preserve $d_k \geq 0$, so raw ascent gives $J(\theta + \eta d) \geq J(\theta)$ for $\eta \geq 0$. Require on-policy scores and omit regularization/momentum.
- Interpret this as an exact benchmark explanation of the protection rule. Multiple actions per stage and shared neural parameters remove the sign guarantee; test those effects instead of claiming the theorem covers the robot unchanged.

C. Conditional Improvement for the Neural Controller

- **Proposition D (actual-update condition).** If expected success has an L-Lipschitz gradient on the update segment,

$$J(\theta + \Delta) - J(\theta) \geq \nabla J^\top \Delta - \tfrac{L}{2}\|\Delta\|^2. \tag{3}$$

- Proof sketch: integrate the gradient along the segment. Positive directional alignment and sufficiently small displacement imply improvement. Apply to the actual optimizer displacement; PPO clipping alone does not prove the premise [7].
- For $d = c\nabla J + e$, $c > 0$, $\|e\| \leq \rho\|\nabla J\|$, $\rho < c$, a sufficient range is $0 < \eta < 2(c-\rho)/[L(c+\rho)^2]$. The current evidence does not bound ρ or L.
- Measure $\nabla J^\top d$ using independent full-trajectory terminal-reward score estimates or paired finite differences. Report uncertainty, update scale, stage interference, and deployment performance; keep this audit independent of the batch choosing d.
- Factor $J = p_{\text{approach}} p_{\text{pickup}|\text{approach}} p_{\text{strict}|\text{pickup}}$ without assuming independence. Use this to explain why later-stage progress can be offset by prefix loss.
- Full proofs, assumptions, counterexamples, and code correspondence are in the internal mathematical note. The main paper will retain only approved, page-budgeted results; no unproved convergence or physical-transfer guarantee.

VI. Experimental Setup

A. Simulation, Protocol, and Baselines

- Describe the five-channel CDPR, MuJoCo Warp/MJ-Lab simulator, cameras, four object categories (apple, orange, potato, tomato), and plate/bowl destinations. Freeze the exact configuration for the selected run.
- Historical matched protocol: 512 development scenes excluding the 128-scene in-run panel, four repeats per checkpoint, 128 decisions, zero settling decisions, no teacher/yaw assistance, deterministic residual mean with prior noise scale 1.
- Use identical scene manifests, horizons, termination rules, and deployment stochasticity across comparison arms. Keep non-finite simulation episodes in denominators and report their rate.
- Core baselines to run or recover: mapped frozen VLA; bootstrapped pre-full-task-RL initializer; same architecture with terminal-only credit; matched continued training with historical versus latent likelihood; latest complete method.
- Prioritize a fixed-prior/no-prior residual control with matched observations and budget to test the VLA action prior's contribution. Feature or language ablations are separate tests; do not interpret prior removal as removal of all VLA information.
- Report successes/trials and scene-paired uncertainty; cluster repeated rollouts by scene. Add independent training seeds when feasible and distinguish rollout variability from training-seed variability.
- Report selected actions, all sampled/executed actions, updates, GPU-hours, and inherited training cost. A 2M pilot continuation is not a 2M-from-scratch method.

B. Tests of the Mathematical Mechanism

- **[Proposed]** compare latent, clipped-sample and independent-marginal scores with exact/finite-difference gradients in controlled processes; then validate recorded robot likelihoods.
- **[Proposed]** compare empirical mixed-group frequency with the per-scene binary prediction; measure losses from stage masks, invalid episodes, and post-selection weighting.
- **[Proposed]** compare protected ($\kappa = 0$) and symmetric ($\kappa = 1$) terminal credit; audit stage-gradient interference, independent full-task directional alignment, and actual update displacement.
- Separate analytic-model verification, stochastic training-policy diagnostics, and deterministic-mean deployment results. Match interaction budgets and report uncertainty for robot comparisons.

C. Physical Robot Protocol (Pending)

- **[Pending: robot configuration, cameras, objects/receptacles, state estimation, control frequency, calibration, and checkpoint]**. Verify every actor input is available on hardware.
- **[Proposed]** evaluate unassisted empty-start pick-and-place with a fixed trial list, randomized execution order, explicit outcome rubric, video records, and baseline under the same hardware conditions.
- **[Proposed]** provisional feasibility design: four objects $\times$ two receptacles $\times$ ten trials = 80 trials per policy, if supported by the actual apparatus. Final sample size follows a precision/power and resource decision before outcomes are reviewed.
- Record every intervention/reset, abort, failure, and adaptation step. Distinguish direct transfer from hardware fine-tuning or recalibration; use an observable physical counterpart of the strict simulator rubric.

VII. Results

A. Full-Task Simulation Performance

- **[Author-reported]** latest run exceeds 50% success and belongs to the latest method direction. Its checkpoint is absent from retained reports. Reserve the headline row; obtain exact successes/trials, checkpoint hash, metric definition, split, and selection rule before claiming strict held-out performance.
- **[Recorded]** older matched development comparison: source 756/2048 (36.91%), latent-likelihood bounded-residual continuation 843/2048 (41.16%), added correction-network arm 769/2048 (37.55%). These are four repeats on 512 scenes, not 2048 independent scenes.
- **[Recorded]** control minus source: +4.25 percentage points, paired 95% CI [+2.10,+6.40], Holm-adjusted $p = 0.00030$. This supports the continued control's gain, not the isolated causal effect of changing likelihood.
- **[Recorded]** older control strict rates: plate 45.43%, bowl 36.86%; physical grasp 76.37%, held lift 68.80%, native placement 47.90%. Use stage retention to explain what changed.
- Table I plan: matched baselines and latest method; columns for strict successes/trials, strict rate with uncertainty, plate/bowl rates, native placement, and interaction budget. Separate unmatched historical results from this table.
- Figure 2 plan: strict validation versus cumulative interaction, with protocol/likelihood changes marked. Distinguish in-run validation from standalone evaluation; no smoothing that hides regression or seed variation.

B. Validation of the Mathematical Mechanism

- **[Recorded]** independent finite-model checks, not robot outcomes: binary-group identities agree with exhaustive enumeration to 1.2×10^{-15} absolute error across $G = 2, 4, 8$ and seven probabilities.
- **[Recorded]** all 336 enumerated sequential-gate groups for $G = 2, 3, 4$ satisfy the protected-credit sign property; three parameter vectors and three raw-ascent step sizes preserve full-task success. These checks complement the proof and do not replace its assumptions.
- **[Recorded]** in a two-decision toy process with persistent offsets, the conditional score agrees with analytic and finite-difference gradients. At $\theta = 0.1$, the exact derivative is 0.3170; scoring independent exact one-step marginals yields 0.6344. This isolates temporal conditioning, without relying on a Gaussian approximation to the marginal.
- **[Pending: CDPR mechanism measurements]**: likelihood checks on actual records, per-scene mixed-group frequencies, protected/symmetric ablation, stage-gradient interference, independent update alignment, and deployment-policy comparison.
- Figure/table plan: theoretical prediction versus measured mechanism, followed by matched full-task improvement. Keep analytic examples visually distinct from simulation and physical success rates.

C. Ablations and Failure Analysis

- Table II plan: staged versus terminal-only credit; full-task bonus on/off and protected/symmetric prefix credit; legacy versus latent likelihood at matched continuation budgets; prior contribution control. Mark all unrun cells pending.
- **[Recorded]** the added correction-network arm has no demonstrated strict gain versus source: +0.63 percentage points, paired CI $[-1.76,+3.03]$. It fails the grasp-retention rule; do not describe that architecture as the successful new method.
- **[Recorded]** non-finite episode rates: source 0.98%, control 1.12%, correction 3.56%. Report beside success and assess paired effects from episode records; missing paired failure intervals in the retained JSON cannot be invented.
- Figure 3 plan: approach/grasp/lift/completion funnel plus scene-position breakdown. Older control's far-side target-y quartile remains at 7.56% strict, motivating spatial robustness tests for the latest policy.
- Audit failures by scene and stage: no grasp, grasp without lift, carry slip, incorrect release/placement, and simulator failure. Define an exclusive hierarchy for a stacked plot; raw overlapping event counters cannot simply be summed.
- Keep the correction-arm result as a concise negative comparison. Earlier SFT ties, retention cycles, and diagnostic trajectories are background unless they answer a specific main-paper hypothesis.

D. Real-World Manipulation

- **[Pending: experiment in progress; no numerical result yet]**. Retain this subsection in the main Results section and fill it only from completed physical trials.
- Table III plan: policy, transfer/adaptation condition, successes/trials, strict success with uncertainty, object/destination breakdown, intervention count, and completion time. Pending entries are not zero successes.
- Figure 4 plan: actual robot setup and a time-ordered successful rollout, paired with a representative failure and measured failure frequency; record camera viewpoints and task instruction.
- Compare the selected policy with the declared hardware baseline on matched conditions; describe observed calibration, cable-dynamics, perception, grasp, and release gaps.
- Conclusions permitted after measurement: feasibility and limits on this apparatus. Do not prewrite successful sim-to-real transfer, $> 50\%$ hardware success, or improvements over baseline.

VIII. Discussion and Limitations

- State precisely which assumptions were tested: conditional sampling, stage separability versus interference, gradient alignment, update size, and exploration-to-deployment mismatch. Theorems explain mechanisms and sufficient conditions; the observed $> 50\%$ score remains an empirical result.
- Interpret staged learning through the measured bottleneck shifts and ablations. Separate observed gains from hypotheses about credit assignment, prior usefulness, and distribution shift.
- Report scope: one target embodiment, limited objects/destinations, manually specified rewards and calibration, inherited pretrained data, and historical scripted assistance.
- Address data/compute cost, checkpoint selection, scene reuse, seed coverage, and numerical simulator failures. Development promotion is not final-test evidence.
- Discuss whether action-prior saturation undermines the expected benefit of a VLA prior; distinguish visual/language feature transfer from action-prior transfer.
- Physical limitations and transfer claims follow completed trials. Outline the next questions only after identifying what the actual hardware evidence supports.

IX. Conclusion

- Restate the approved contribution: staged residual CDPR learning with a mathematically specified likelihood and protected credit rule, supported by explicit propositions and mechanism tests.
- Insert the latest verified full-task result and its comparison once available; include the physical result only after completion.
- Close with the measured scope and the main remaining failure mode. Avoid universal transfer, convergence, or state-of-the-art claims without supporting experiments.

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